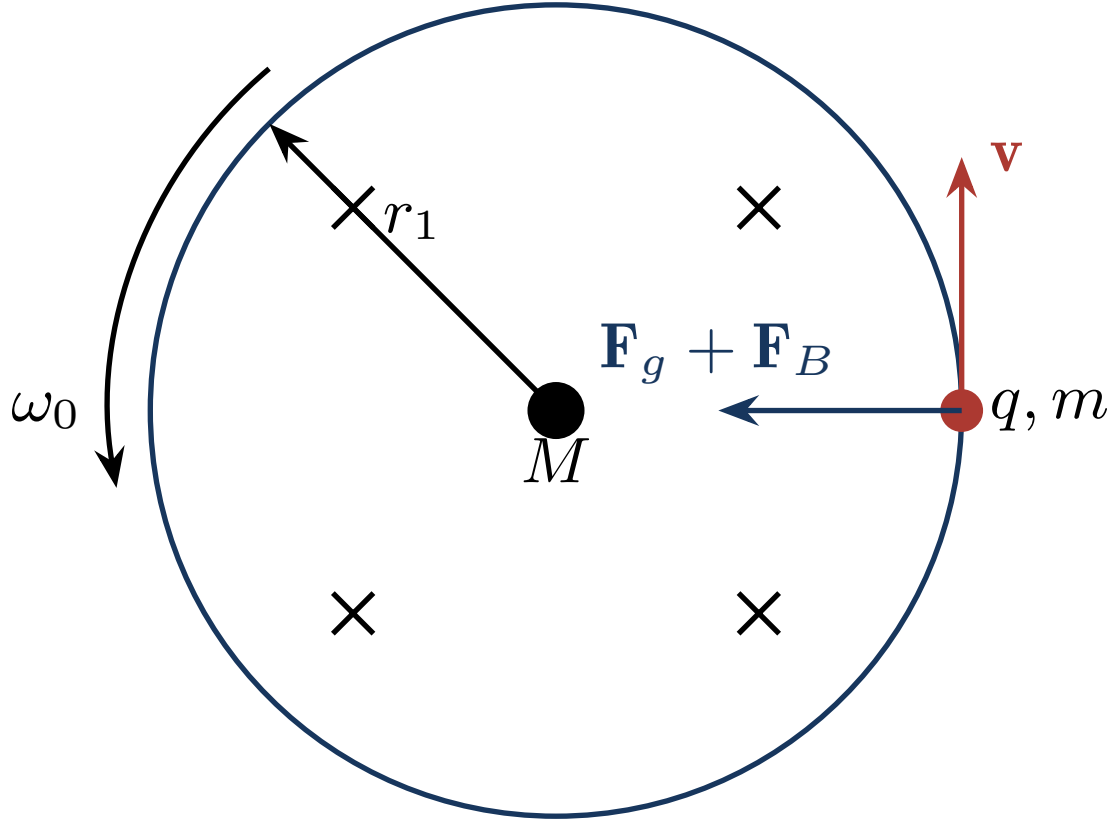
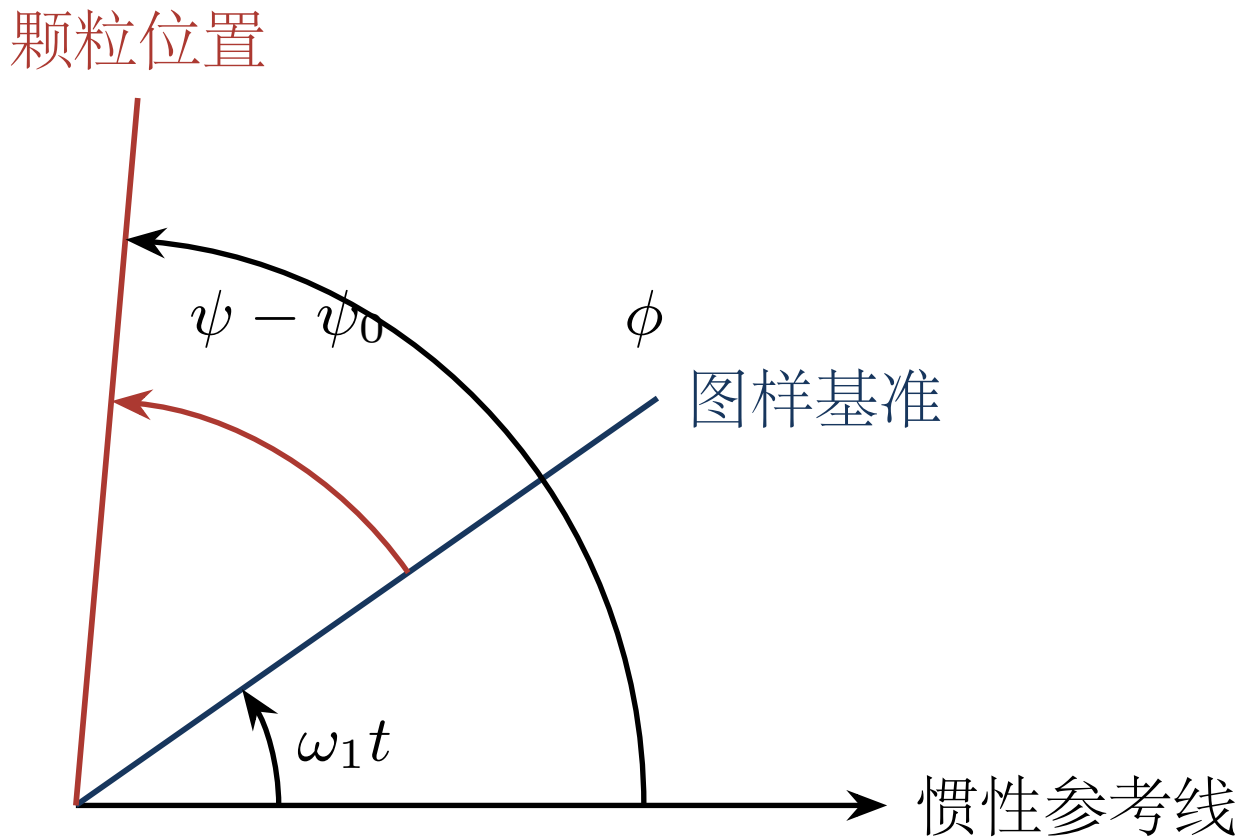
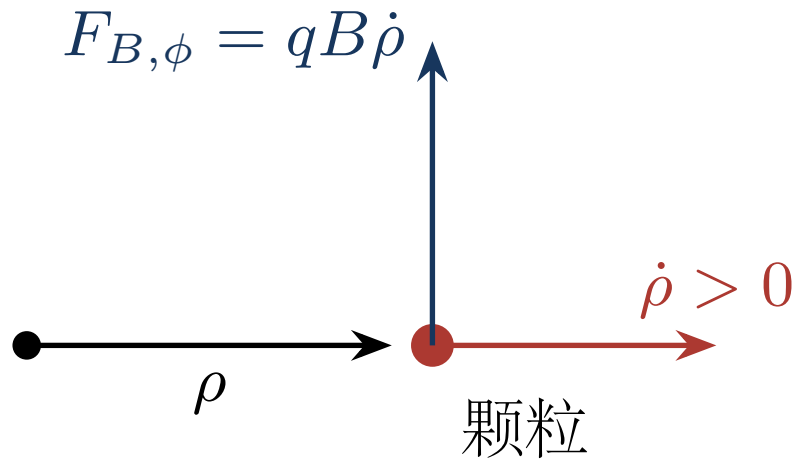


北极视角：$\mathbf{B} = -B\hat{\mathbf{z}}$，正电荷磁力向内



$$\psi = \phi - \omega_1 t + \psi_0$$

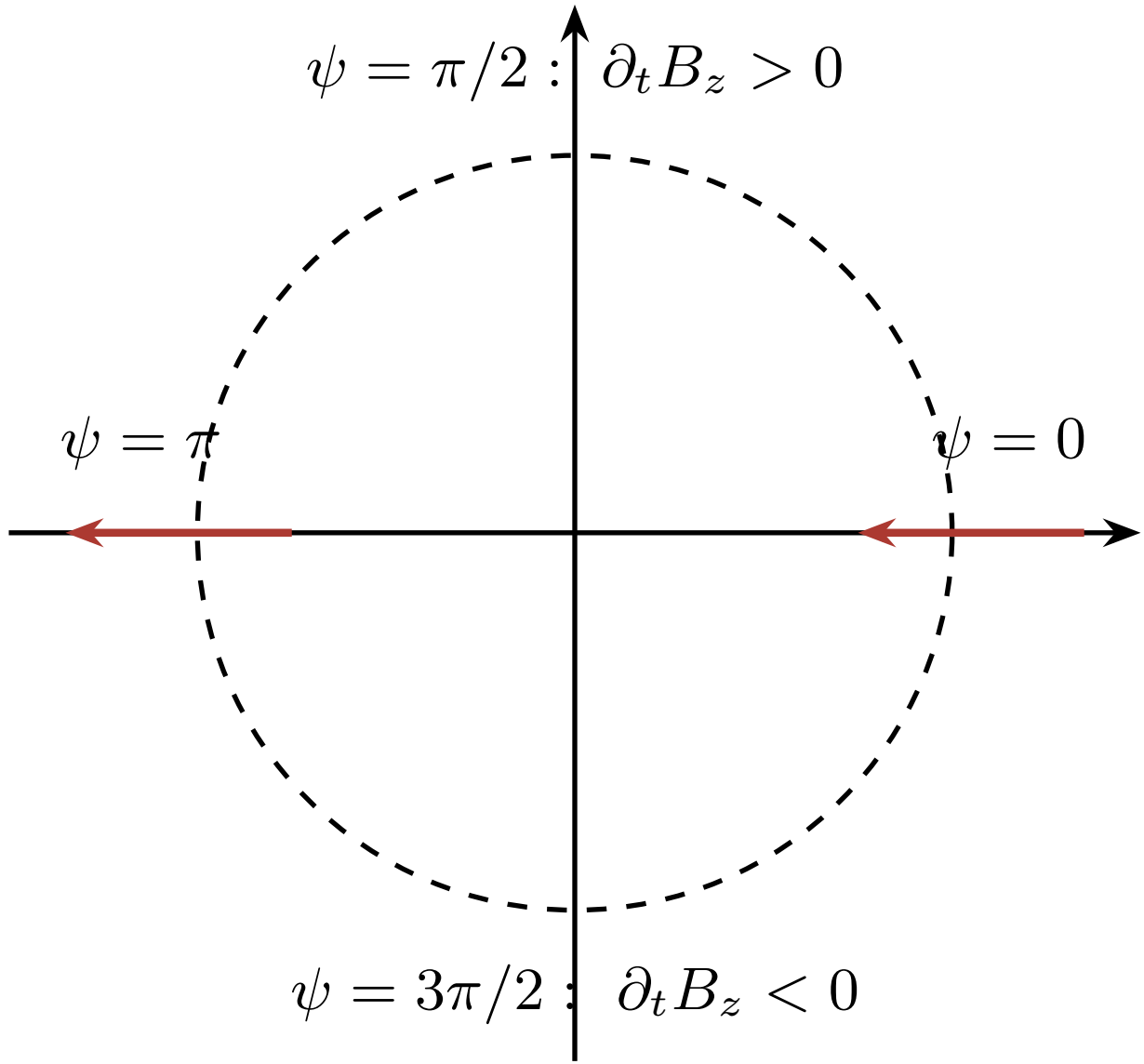
沿颗粒：$\dot{\psi} \simeq \omega_0 - \omega_1$



$$\dot{L} = qB\rho\dot{\rho}$$

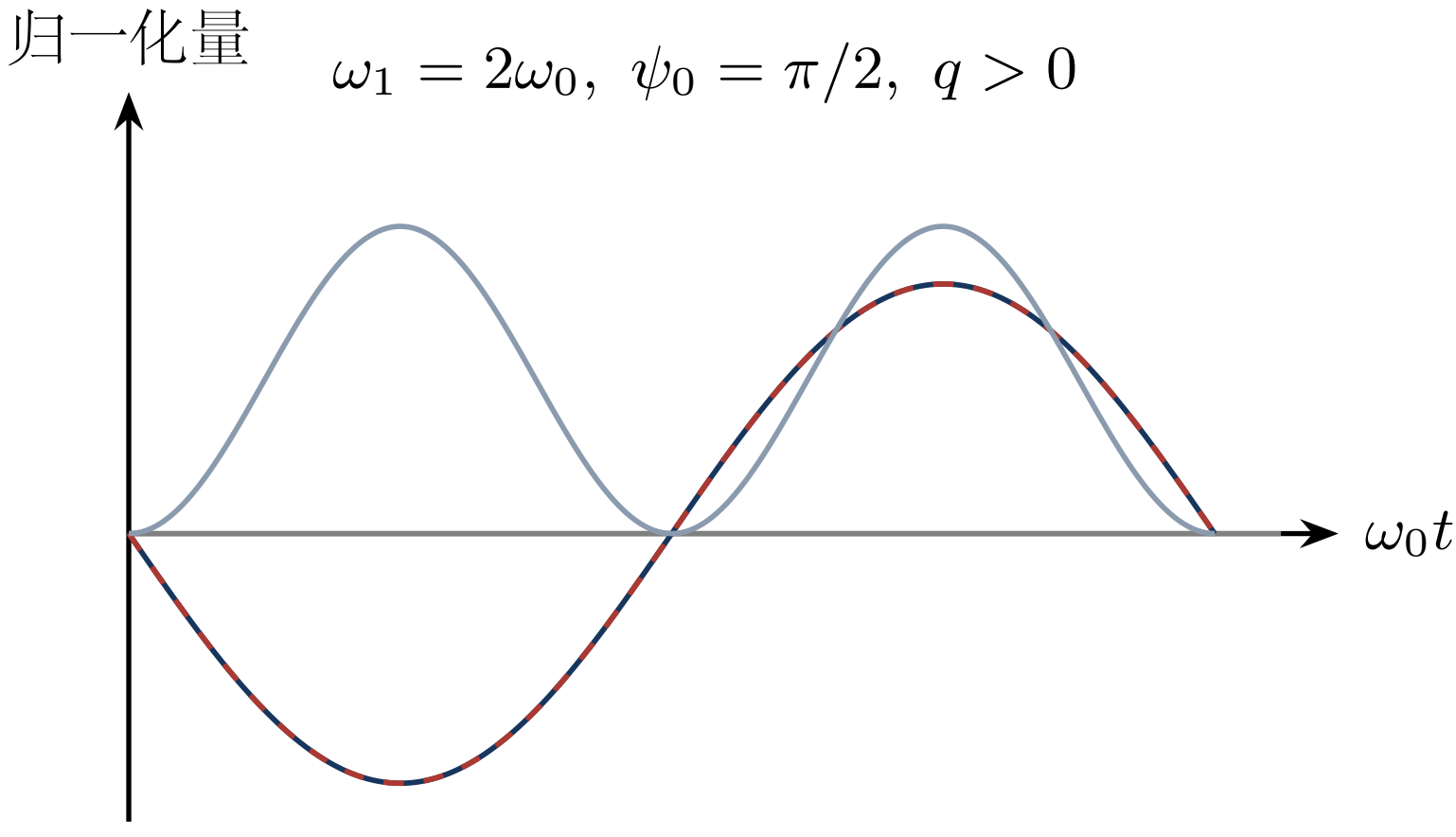
$$J = L - qB\rho^2/2$$

机械角动量可变；磁力仍不做功



$$E_r = -r_1 \omega_1 \Delta B \cos\psi$$

图示取 $\omega_1 \Delta B > 0$；电场箭头分别向内、向外



蓝实线：径向速度；红虚线：同相电场（重合）

浅蓝：瞬时功率非负，整周期平均为正